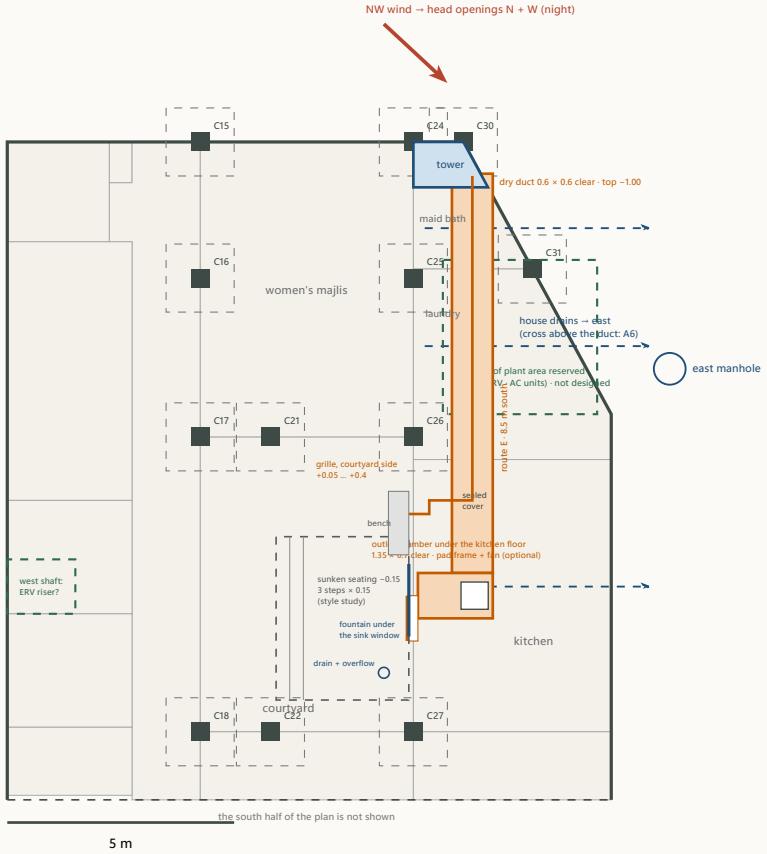
Air system study v2 · dry malqaf for night ventilation, small dry duct, wall outlet · Taif / Palm House V6.3

design-development

study · every size and performance statement ASSUMED / NOT CALCULATED, to be designed by the MEP engineer · CSV coordinates, north up

Nothing here is calculated: no airflow, cooling, filtration, structural or hydraulic figure. The study proves the performance of no option. ALL CHECKS PASSED means the drawings are consistent with the project files, not that the system was sized. v1 (air-p1, walk-in tunnel with a water channel) is withdrawn.

Plan at foundation level · tower, duct (route E), chamber, grille · pads dashed · reserves green



1 · The main ventilation system (the base, independent of the malqaf)

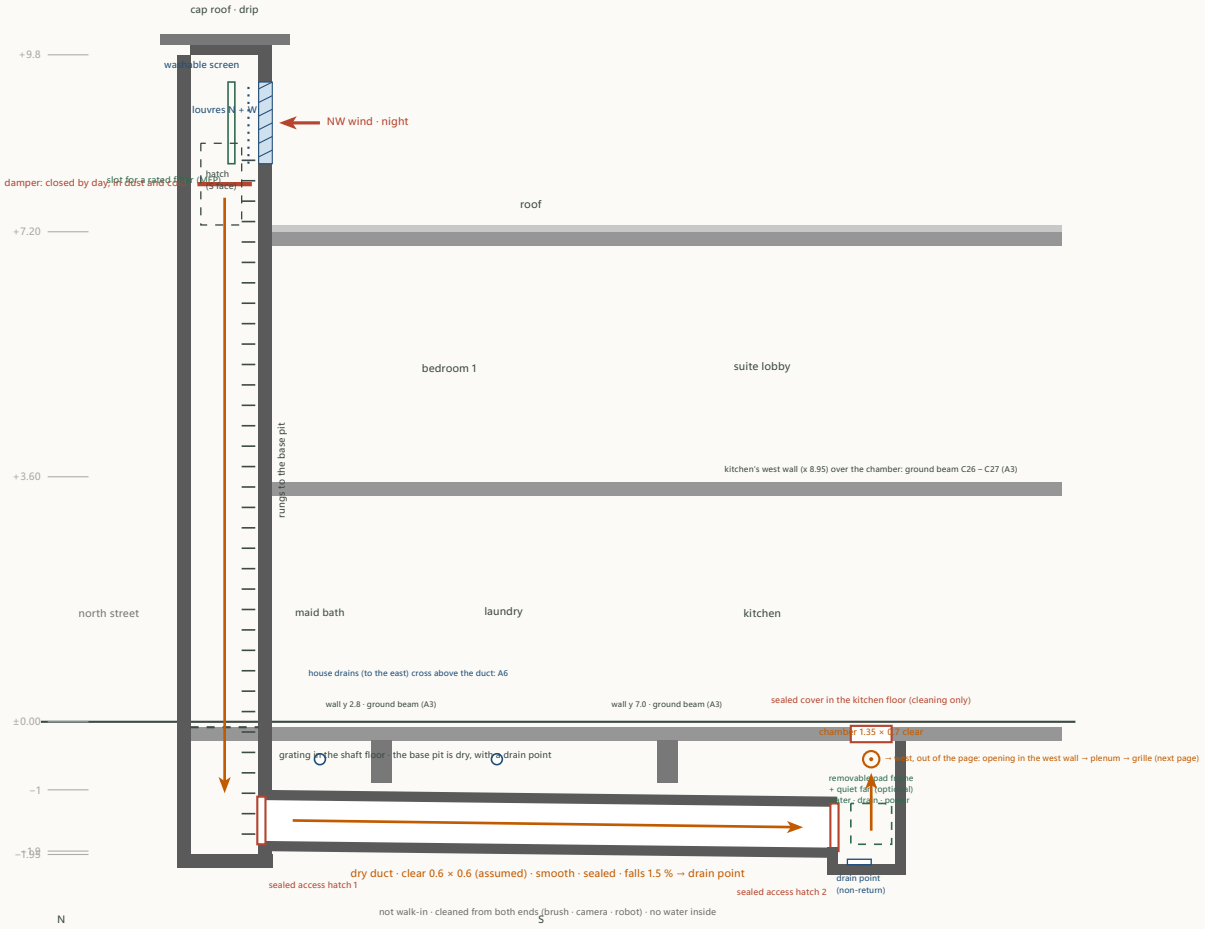
- The house needs a **calculated main system, designed by the MEP engineer**: **filtered fresh air to the closed rooms**, above all the bedrooms, through an ERV / heat-recovery unit; **exhaust** from the bathrooms, the WCs and the kitchen (hood and extract through the east wall); **air-conditioning sized separately**, with the ducts and the AC in the gypsum ceilings (the owner's rule: drop beams, ducts and lighting hidden by gypsum ceilings).
- **The malqaf is a natural supplement, not the main system.** An outlet into the open courtyard does not ventilate the closed rooms; only the ducted system does.
- **Reserved, not designed** (open items M1–M3): a **plant area on the roof** over the maid wing (dashed green, ≈ 11.56 m², screened, beside the tower, away from the suite court and the bedrooms' windows) for the ERV and the AC outdoor units; **duct routes in the gypsum ceilings** of both floors; a **riser** (the west shaft, 1.5 × 1.2, dashed green, or a second one beside the east kitchen stack) between the floors and the roof plant.

2 · The malqaf, simplified: dry, for night ventilation

- **Why v1 was withdrawn:** water inside a dark underground duct adds condensation, mould and bacteria risks; a short, shallow tunnel gives little ground cooling; a walk-in tunnel (deep excavation, waterproofing, pumps, footings) is too complex for a house; passing air over water does not clean it, and a screen is not a rated filter.
- **Now:** the tower stays **dry**; a washable insect / dust screen **plus a slot for a proper rated filter** (MEP: class and pressure drop); a **closable damper** (closed in dust storms, cold weather and the hot daytime hours; motorised with a manual override); a roof access hatch. **Main use: night ventilation** — Taif's cool nights: open at night to cool the thermal mass of the courtyard walls and floors, closed by day. **No water anywhere inside the tower or the duct.**
- **Duct:** a short, small, **dry** duct from the tower base **south under the maid bath and the laundry** (route E, 8.5 m on x 10.25) to a chamber under the kitchen floor, assumed **0.6 × 0.6 m clear** (to be sized), smooth, sloped 1.5 % to a drain point, **sealed against soil gas and ground water**; just under the assumed tie beams (top −1), 0.10 m from the assumed pads; the maid bath and laundry drains cross **above** it (A6); **not walk-in**: cleaned from both ends through **sealed hatches** at the tower base and at the chamber.
- **Outlet in the wall, not in the floor (owner, 3d-char2):** a low grille at the **north end of the courtyard's east wall** (the kitchen's west wall, courtyard side; V6.4: the wall's first 0.5 m, y 6.5–7.0, belongs to the laundry, the grille at y 10–11 stays in the kitchen part over the chamber; +0.05 … +0.4), mashrabiya-type, **discharging into the courtyard just above the bench of the sunken seating** (floor −0.15, style study): the cool air settles there; the plenum stays below counter level on the kitchen side; the women's majlis glass stays complete (A7 resolved). **No air opening in the courtyard floor**; the cleaning access is a **sealed cover in the kitchen floor**.

Long section along the air path · tower head → shaft → base pit → dry duct (south, x 10.25) → outlet chamber under the kitchen floor → (west) plenum → grille → sunken seating

along the duct axis, north → south; at the chamber the path turns west into the kitchen's west wall (the section looking north on the next page) · levels from the finished ground floor ±0.00 · all assumed / not calculated



How it works (qualitative only; nothing calculated)

- **Night ventilation:** in the evening the damper opens; the head catches the north-west breeze above the roof (or the cool, heavy night air simply sinks down the shaft); the air passes the screen and the filter, falls down the shaft, runs through the short dry duct and leaves slowly through the low grille in the kitchen's west wall, just above the bench of the sunken seating, and settles into the sunken area, cooling the courtyard's walls and floor. In the morning the damper closes and the cooled mass keeps the courtyard pleasant longer. **How much:** not calculated; the MEP engineer estimates the flow (wind-driven and stack) and decides whether a small fan is needed (option B).
- **Hot daytime hours:** the damper is closed. If the owner wants cooled supply by day, the **optional pad frame and quiet fan** in the outlet chamber are installed for the season (option B): the fan draws through the pads; the pads are **drained and dried daily**; the tower and the duct stay dry.
- **The closed rooms** (bedrooms, baths, kitchen) are ventilated and cooled by the **main system** (ERV, exhausts, AC), not by the malqaf.

Health and dryness rules (stated, not optional)

- **No water inside the tower or the duct:** no channel, no sprays, no mist, no fogging. The only water is in the optional pad frame in the outlet chamber, drained and dried daily; make-up water through a float valve and a backflow preventer.
- **Filtered:** a rated filter in the head's slot (class and pressure drop by the MEP engineer); the screen only keeps insects and coarse dust out.
- **Sealed and drained:** the duct is sealed against soil gas and ground water, falls to a drain point with a non-return trap; the base pit and the chamber have drain points; condensation, if any, runs to them.
- **Separate from the house and closable:** a separate element; no house drain crosses it; the damper closes it; the floor cover is gas-tight and opened only for cleaning.

Levels along the path (assumed)

head top / head openings	+9.8 / +8.2 … +9.4
roof finish / roof slab	+7.30 / +7.20
ground floor finished / grating in the shaft	±0.00 / −0.08
tower base pit floor	−1.95
duct top / underside at the tower	−1 / −1.9
duct floor (clear) at the tower / at the chamber	−1.75 / −1.85
outlet chamber floor	−2.10
grille in the kitchen's west wall, courtyard side	+0.05 … +0.4
sunken seating floor / bench seat (style study)	−0.15 / −0.05 (the grille 0.60 / 0.20 above them)
assumed tie beams / pad footing underside	−0.28 … −0.90 (the duct passes 0.10 m under them) / −1.50 (only the tower pit and the chamber reach it: A1, A3)

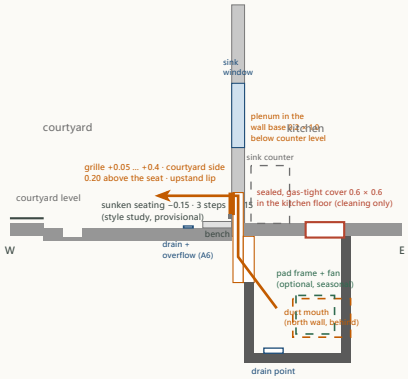
Duct cross section · outlet chamber and grille · the three options for the MEP engineer

all sizes assumed · the duct, the tower base pit and the chamber by the structural engineer; filter, fan, pads, water, drain and power points by the MEP engineer

Duct cross section (at a tie-beam crossing)



Outlet · section, looking north



Outlet chamber and grille (assumed)

- **Chamber (owner, 3d-char2):** under the kitchen floor against the kitchen's west wall (x 8.95), outer x 9.05–10.7, y 9.5–10.5 clear 1.35 × 0.7, floor −2.10, walls 0.15, sealed like the duct; 0.15 m from the assumed C26 pad, under the sink counter. The duct enters through its north wall; a **sealed access hatch** there for cleaning the duct from this end.
- **Air path:** chamber → an opening in its west wall → a **plenum in the wall base** (0.2 × 1.0, y 10–11, from the slab to the grille head, below counter level on the kitchen side) → the **grille** on the courtyard face, +0.05 … +0.4, 1.0 m wide, decorative (mashrabiya-type), with an upstand lip, an insect screen and a closable shutter behind it; **the air discharges into the courtyard, never into the kitchen**.
- **Sunken seating (style study, 3d/character.yaml):** the grille discharges just above the built-in bench (seat −0.05) of a sunken area −0.15 deep (3 steps × 0.15 from the west) against the east wall, x 5.93–8.85, y 8.7–12.3 level edges ≥ 1.0 m at every door and glass wall; a **floor drain at its lowest point plus an overflow** (A6); a wall fountain under the sink window, decorative (A8 / A6).
- **Optional, installed only for the hot season:** a removable evaporative pad frame (≈ 0.6 × 0.6) and a small quiet fan in the chamber, with water, drain and power points. Rules: no sprays or mist; pads drained and dried daily; the duct and the tower stay dry.
- **Access:** a sealed, gas-tight cover 0.6 × 0.6 in the kitchen floor east of the sink counter (x 10–10.6), opened only for cleaning and for the seasonal pad frame; **not an air opening**. The chamber is 2 m deep: a short ladder stored nearby, two people, lid open (A5).
- **The glass:** the women's majlis glass (x 5.1–8.1) on the north wall stays complete: A7 is resolved by moving the outlet to the east wall.

The three options the MEP engineer compares before anything goes into the foundations

	A · direct dry malqaf	B · malqaf + fan + filter	C · dry earth-air heat exchanger
what	tower, screen + filter, damper, the short dry duct, the wall grille; wind and stack only	as A, plus a small quiet fan (in the chamber or the shaft) and a rated filter; the pad frame optional at the outlet	a longer, deeper dry pipe run (proper depth and length, several parallel pipes), condensate drainage, a fan; the courtyard grille or ducted supply
gives	night ventilation of the courtyard when the breeze blows; nothing by day	controlled night ventilation whatever the wind; filtered; optional evaporative supply by day	cooled (and in winter tempered) supply by day and night; the only option with real ground cooling
costs / risks	lowest; depends on the wind; the courtyard only	fan power, filter changes; the pads need the daily drain-and-dry routine	excavation outside the house, length, condensate and hygiene management, fan power; must be designed, not assumed
to compare	annual performance (hours of useful supply, temperatures), capital and running cost, maintenance	idem	idem

The study proves the performance of none of them. The plan and the sections show option A (the drawings also fit B); C would change the route and the depth and is not drawn.

Route: the owner's choice (3d-char2) against the two candidates of v1 / v2

	E · to the kitchen's west wall (chosen, owner)	A · under the women's majlis (v2, superseded)	B · under the maid wing to the kitchen north wall (v1, rejected)
path	tower base → 8.5 m south on x 10.25 → chamber under the kitchen floor → west into the kitchen's west wall → grille over the sunken seating	tower base → 1.68 m west → 5.85 m south on x 7.27 → chamber at the courtyard's north wall	south under the maid bath and the laundry, west under the kitchen north wall; ≈ 8.5 m
above	maid bath, laundry, kitchen	the women's majlis only	maid bath, laundry, kitchen wall
footings	the tower base pit (A1); ground beams under the walls y 2.8, y 7.0 (V6.4) and the kitchen's west wall (A3); 0.10 m from the C25 / C26 / C31 pads	the majlis south wall ground beam; 0.48 m from C25, 0.4 / 0.25 m from C21 / C26	walls y 1.0 and 2.8, the RC wall W5, the C26 pad
drains	the maid bath and laundry drains cross **above** the duct: clearance and protection (A6)	none cross it	crosses the maid bath and laundry drains
why	the cool air settles where people sit; the women's majlis glass stays complete (A7 resolved)	the grille cut the women's majlis glass (A7)	—

Maintenance, reserved spaces for the main system, open items

Maintenance table (what · how often · how to reach it)

Item	What to do	How often	How to reach it
head louvres + insect / dust screen	brush and wash the screen; clear the louvres; check the drip	monthly in the dust season, else every 3 months	from the roof, through the hatch in the head's south face
rated filter (head slot)	check the pressure drop / the indicator; replace or wash as its class allows	per the MEP engineer; before and after the dust season	from the roof hatch
damper (head)	check the motor, the manual override and the control; grease, test closing	every 6 months and before the dust season	from the roof hatch
shaft walls, rungs, grating	brush, check the rungs, the lighting and the grating lock	yearly	rungs from the head hatch
tower base pit + drain point	sweep, check the drain and its trap, look for damp	every 6 months	down the shaft (two people, lid open)
dry duct	brush / camera inspection / cleaning robot from both ends; check that it is dry and sealed	yearly; after any smell or damp	sealed hatch 1 (base pit), sealed hatch 2 (chamber)
outlet chamber + drain point	sweep, check the drain and the trap, the water and power points	every 6 months	sealed cover in the kitchen floor (cleaning only)
grille, screen and shutter	wash the screen, clear the grille and its upstand, test the shutter	monthly in use	from the sunken seating, standing on the bench
sunken seating drain + overflow	clear the grate and the overflow; check after rain	monthly; after every rain	in the courtyard
pad frame + fan (optional, seasonal)	install for the season; pads drained and dried daily; clean the pads, the tray and the fan	daily drain (automatic); pads monthly; in / out of the season	through the floor cover
sealing, joints	look for damp, cracks, salts, smells (soil gas)	after every heavy rain; yearly	hatches and cover
main system (ERV, exhausts, AC)	filters, coils, condensate, fans, controls	per the MEP engineer's schedule	roof plant area; ceiling access panels

Reserved for the main system (not designed, open items M1–M3)

- **Roof plant area** over the maid wing, x 9.6–13, y 2.6–6 (≈ 11.56 m²): ERV unit, AC outdoor units, screened from the street and the suite court; a roof hatch or stair for service; structure to allow for the loads (engineer).
- **Risers:** the west shaft (1.5 × 1.2, x 0–1.5, y 9.2–10.4) rises from the utility room to the roof beside the bedrooms; a second riser beside the east kitchen stack if the MEP engineer needs it.
- **Duct routes in the gypsum ceilings** of both floors (the owner's rule), with access panels; the first-floor rooms over the kitchen and the men's majlis have ribbed slabs: the ducts run below them in the ceilings, not through them.

Open items for the engineers

M1	MEP	**The main ventilation system:** filtered fresh air to the closed rooms (ERV / heat recovery, the bedrooms first), exhausts for the baths, WCs and the kitchen, the AC sized separately; the design, the plant location (the reserved roof area) and the duct routes in the gypsum ceilings.
M2	MEP / structure	The roof plant area: equipment loads, screening, vibration and noise to the bedrooms and the suite court, service access.
M3	MEP	The risers (the west shaft; a second one if needed) and the ceiling heights with the ducts (the gypsum ceilings at ≈ +3.05 … +3.20 under the drop beams).
A1	structure	The tower base pit (−0.08 … −1.95) with the shaft walls as the foundation of the tower and the ends of W3 / W4; the duct leaves it through the W3 line: design the pit and the crossing.
A2	structure / MEP	The duct: a factory-sealed box (precast RC or GRP) or an in-situ sealed box; its size (0.6 × 0.6 assumed), joints, bedding, soil gas and ground-water sealing; soil report first (X9).
A3	structure	The walls over the duct: the maid bath / laundry wall (y 2.8) and the laundry / kitchen partition (y 7.0, V6.4) on ground beams bridging it; the kitchen's west wall (x 8.95) over the chamber on a ground beam between the C26 and C27 pads, with the plenum in its base.
A4	MEP	Option study A / B / C: annual performance, cost and maintenance; the head opening sizes, the filter class, the fan, the damper control (night / day, dust, cold).
A5	MEP / safety	Access: the shaft rungs and fall protection; the base pit and the chamber as confined spaces (two people, lid open); the sealed hatches and the floor cover gas-tight and lockable.
A6	MEP	The drain points of the pit, the duct and the chamber (traps, non-return, where they discharge); the maid bath and laundry drains crossing above the duct (vertical clearance, protection); the sunken seating's floor drain and overflow at its lowest point (open to the sky: size them for the rain), and the fountain's drain.
A7	owner / design	**Resolved (3d-char2):** the outlet moved to the courtyard's east wall; the women's majlis glass (x 5.1–8.1) stays complete.
A8	MEP	The optional pad frame and fan at the outlet: water quality (float valve, backflow preventer, daily drain-and-dry), noise, power; no sprays or mist (rule).

Sizes assumed: head 2.5 m above the roof, openings 0.8 × 1.2 on the north and west faces; shaft ≈ 0.78 × 1.3 clear; base pit to −1.95; duct clear 0.6 × 0.6 in a 0.9 box, top −1.00, fall 1.5 %; chamber 1.35 × 0.7 clear under the kitchen floor, floor −2.10; grille 1.0 wide in the kitchen's west wall, +0.15 … +0.75 (sunken seating −0.45, bench seat −0.05: style study); roof plant area ≈ 11.6 m²; pads 1.5 × 1.5 with underside −1.50, tie beams to −0.90. Nothing here replaces the MEP or structural design.

✓ Air system study v2.1 · dry malqaf · route E · duct 8.5 m · outlet in the kitchen's west wall over the sunken seating · main system by the MEP engineer · 10 open items (A7 resolved)